Escape and the Diagonal

The games D(m, r), solved for every m ≥ 3 except m = 4, with the two diseased moduli explained.

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The game. D(m, r) is played on the non-negative integers. From a position $n \equiv r \pmod{m}$ — *squares mode* — a move subtracts any square k^2 with $k \geq 1$ and $k^2 \leq n$. From any other position — *fallback mode* — the only move subtracts m, available when $n \geq m$. Normal play: a position with no legal move is P; a position is N iff some move reaches a P. Throughout, Q(2m) denotes the set of squares modulo 2m.

Part I — The engine: squares modulo 10

k	k²	k² mod 10
1	1	1
2	4	4
3	9	9
4	16	6
5	25	5
6	36	6
7	49	9
8	64	4
9	81	1
10	100	0

Collecting the distinct values: **Q = {0, 1, 4, 5, 6, 9}**. The complement — the values a square can never take mod 10 — is {2, 3, 7, 8}.

Why k = 10 is enough, one line and part of the proof: $(k + 10)^2 = k^2 + 20k + 100$, and both 20k and 100 are multiples of 10, so $(k + 10)^2 \equiv k^2 \pmod{10}$ — every k past 10 replays one of these ten rows, so the table above is the whole story forever.

Mirror check, kept as the standing tool for every later table: $(10 - k)^2 = 100 - 20k + k^2 \equiv k^2 \pmod{10}$ for the same reason. The table must mirror around the middle, and it does: row 6 matches row 4 (both 6), row 7 matches row 3 (both 9), row 8 matches row 2 (both 4), row 9 matches row 1 (both 1). Footnote kept honest: row 10 mirrors "row 0," so the lone 0 enters at that end — k = 1..5 catches {1, 4, 5, 6, 9} and k = 10 supplies the 0. A broken mirror means an arithmetic slip.

Part II — The Escape Theorem for D(5, 2)

II.1 Statement. There is an explicit threshold N_0, computed in II.6 below, such that every position $n \equiv 2 \pmod{5}$ with $n \geq N_0$ has at least one legal square move — some k with $k^2 \leq n$ — whose landing $n - k^2$ is a P-position of a fallback chain. Consequently every such n is an N-position.

II.2 The target window. By the Chain Lemma, the fallback classes 0, 1, 3, 4 (mod 5) have their P-positions exactly at $n \equiv 0, 1, 3, 4 \pmod{10}$; call that set T = {0, 1, 3, 4} (mod 10). A landing in T is genuinely P: a number ≡ 0, 1, 3, or 4 (mod 10) is ≡ 0, 1, 3, or 4 (mod 5), so it lives in a fallback chain where the lemma has jurisdiction, and its residue mod 10 matching the chain bottom's means it sits an even number of rungs up — precisely the lemma's P-rungs.

II.3 The split. Write $n = 5q + 2$: if $q = 2t$ then $n = 10t + 2$, and if $q = 2t + 1$ then $n = 10t + 7$, so a class-2 position is $\equiv 2$ or $7 \pmod{10}$ — that is, r or $r + m$ — and the proof forks on the parity of q exactly the way the chain proof forked on the parity of j.

II.4 Coverage line one. For $n \equiv 2 \pmod{10}$: the differences $2 - T$ are $\{2-0, 2-1, 2-3, 2-4\} \equiv \{2, 1, 9, 8\} \pmod{10}$, and intersecting with $Q = \{0, 1, 4, 5, 6, 9\}$ leaves $\{1, 9\}$; take the smallest, $s = 1$, so subtracting 1^2 gives $n - 1 \equiv 1 \pmod{10} \in T$.

II.5 Coverage line two. For $n \equiv 7 \pmod{10}$: the differences $7 - T$ are $\{7, 6, 4, 3\} \pmod{10}$, intersecting with Q leaves $\{4, 6\}$; take $s = 4$, so subtracting 2^2 gives $n - 4 \equiv 3 \pmod{10} \in T$. Both keystones came with a spare — $9 = 3^2$ also rescues residue 2, and $16 = 4^2$ also rescues residue 7. The corridor is wider than it needs to be.

II.6 Legality and the threshold. The rescue squares are the concrete numbers 1 and 4. For the residue-2 class, whose members are 2, 12, 22, …, legality demands $1 \leq n$, which already holds at the smallest member. For the residue-7 class, members 7, 17, 27, …, legality demands $4 \leq n$, which holds at 7. So both rescues are legal for every member of their class, and the threshold collapses: $\mathbf{N_0 = 2}$, the smallest class-2 position in existence. Below N_0 there are no class-2 positions at all, so the hand-check zone is empty — verdict: vacuous. Checking the workbook rows anyway, as ordered: $2 - 1 = 1$ (P) ✓, $7 - 4 = 3$ (P) ✓, $12 - 1 = 11$ (P) ✓, $17 - 4 = 13$ (P) ✓, $22 - 1 = 21$ (P) ✓, $27 - 4 = 23$ (P) ✓. Every row is rescued-anyway, rebels: none. Worth recording why it collapsed: the rescue squares 1 and 4 happen to be smaller than the smallest members of their classes. Nothing guarantees that luck at other moduli, which is exactly why the theorem was written with N_0 as a named unknown.

II.7 Assembly. The Chain Lemma classifies the four fallback classes everywhere: P exactly at $n \equiv 0, 1, 3, 4 \pmod{10}$, N at 5, 6, 8, 9. The Escape Theorem classifies class 2 everywhere, since $N_0 = 2$ means "past the threshold" is all of it: residues 2 and 7 (mod 10) are all N. Union the two and every residue mod 10 is accounted for, with no unexplained positions beneath any threshold.

Theorem (D(5,2), solved). A position n is P if and only if $n \equiv 0, 1, 3$, or $4 \pmod{10}$.

One coincidence too pretty not to flag: the P-set of the whole game *is* the target window T — the landing pads and the answer turned out to be the same four residues. [Resolved: Part IV, Remark 2.]

II.8 Scope and the forward hook. The proof secretly depended on squares mod 10 being *rich*: Q fills six of the ten residues, enough that both difference lists $\{2, 1, 9, 8\}$ and $\{7, 6, 4, 3\}$ caught at least one member — had Q missed either list entirely, one residue of class 2 would have had no escape and the whole corridor would have collapsed. Generalizing to $D(m, r)$ therefore means asking whether that richness survives at other moduli — whether squares mod $2m$ keep intersecting the difference sets $\rho - T$ for both $\rho \in \{r, r + m\}$ — and since quadratic residues thin out and organize themselves as the modulus grows, that question, not the game logic, is the real content of what follows.

II.9 Verification against the workbook. Prediction first, then the rows: residue-2 positions (2, 12, 22) should be rescued by any legal square $\equiv 1$ or $9 \pmod{10}$, residue-7 positions (7, 17) by any square $\equiv 4$ or $6 \pmod{10}$, and every other square should be wasted.

- $n = 2$: only move $2-1 = 1$ (P). Predicted rescue 1^2 present ✓, nothing else in the row to waste.
- $n = 7$: $7-1 = 6$ (N), $7-4 = 3$ (P). Rescue 2^2 present ✓; the 1^2 is wasted exactly as the congruence says ($1 \notin \{4, 6\}$ mod 10, landing $\equiv 6$).
- $n = 12$: $12-1 = 11$ (P), $12-4 = 8$ (N), $12-9 = 3$ (P). Primary rescue 1^2 ✓, and the spare $9 = 3^2$ makes its first appearance in the first row big enough to afford it.
- $n = 17$: $17-1 = 16$ (N), $17-4 = 13$ (P), $17-9 = 8$ (N), $17-16 = 1$ (P). Primary 2^2 ✓ and spare 4^2 ✓.
- $n = 22$: $22-1 = 21$ (P), $22-4 = 18$ (N), $22-9 = 13$ (P), $22-16 = 6$ (N). Primary 1^2 ✓, spare 3^2 ✓.

The check came back stronger than required: not only does the predicted rescue sit in every row, every wasted square is wasted for the predicted reason — its residue misses the survivor set, and every wasted landing is $\equiv 6$ or $8 \pmod{10}$, both N-residues. The congruence machinery reproduces the workbook letter-for-letter, and the 200,000 mechanical sweep stands as the bulk consistency evidence behind it.

Part III — The engines at every modulus: Q(2m) with complements

The mirror tool, proved once in general: $(2m - k)^2 = 4m^2 - 4mk + k^2 \equiv k^2 \pmod{2m}$, since $4m^2 = 2m \cdot 2m$ and $4mk = 2k \cdot 2m$. So every table is $k = 1..m$ plus mirrors, plus the 0 from $k = 2m$. Two side facts fell out of the mirrors and are used below: for odd m, $m^2 - m = m(m - 1)$ has the even factor $m - 1$, so $m^2 \equiv m \pmod{2m}$; for even m, $m^2 = (m/2) \cdot 2m \equiv 0 \pmod{2m}$.

mod	Q(2m)	complement
4	{0, 1}	{2, 3}
6	{0, 1, 3, 4}	{2, 5}
8	{0, 1, 4}	{2, 3, 5, 6, 7}
10	{0, 1, 4, 5, 6, 9}	{2, 3, 7, 8}
12	{0, 1, 4, 9}	{2, 3, 5, 6, 7, 8, 10, 11}
14	{0, 1, 2, 4, 7, 8, 9, 11}	{3, 5, 6, 10, 12, 13}
18	{0, 1, 4, 7, 9, 10, 13, 16}	{2, 3, 5, 6, 8, 11, 12, 14, 15, 17}
22	{0, 1, 3, 4, 5, 9, 11, 12, 14, 15, 16, 20}	{2, 6, 7, 8, 10, 13, 17, 18, 19, 21}

The mod-4 and mod-8 rows are the indictment written out: Q(4) = {0, 1} and Q(8) = {0, 1, 4} both sit entirely inside the triple {0, 1, m}, so their complements swallow every value any hard set will ever contain (Part IV) — the disease of $m = 2$ and $m = 4$ is visible in these two rows before any game logic starts. One eyebrow kept raised on purpose: the thinnest *odd* table is mod 18, thinned because 9 is itself a square and the rows $k = 3$ and $k = 9$ collide there — $m = 9$'s anomaly is foreshadowed in its own table.

Part IV — The general theorem

Theorem (Diagonal Law). Let $m \geq 3$ with $m \neq 4$, and let $0 \leq r \leq m - 1$. Define the exception set

E = {(9, 4), (9, 5), (9, 6)}.

In D(m, r), a position n is P **if and only if** $n \equiv b \pmod{2m}$ for some foreign bottom $b \in \{0, 1, \dots, m-1\}$ $\{r\}$, **or** $n = 0$, **or** $(m, r) \in E$ and $n = m + r$.

The law holds for both parities of m, with no thresholds and no unverified zones.

IV.1 Setup. By the Chain Lemma (general form, as accepted): fallback moves preserve class mod m, so each class $c \neq r$ is a single self-contained chain $c, c + m, c + 2m, \dots$ with terminal bottom c — no square moves ($c \not\equiv r$) and no affordable fallback ($c < m$) — and P/N alternate upward from the stuck bottom: P exactly at $n \equiv c \pmod{2m}$. The window T(m, r) is the set of foreign bottoms mod 2m. Class r splits mod 2m into the **free residue** (r when $r \geq 1$, else m) and the **hard residue** ($r + m$ when $r \geq 1$, else 0). The free escape is immediate: 1^2 lands the free residue on the bottom directly beneath it ($r - 1$, or $m - 1$ when $r = 0$), and is legal from the smallest free member. For the hard residue, a square value s (mod 2m) lands in the window iff $s \in H(m, r) = \{r+1, \dots, m-1\} \cup \{m+1, \dots, m+r\}$ for $r \geq 1$, or $\{m+1, \dots, 2m-1\}$ for $r = 0$. Constitutional uselessness, for every m and r: $0, 1, m \notin H(m, r)$ — a square $\equiv 0$ or $m \pmod{2m}$ is $\equiv 0 \pmod m$ and stays inside class r, and subtracting $\equiv 1$ from the hard residue lands one below it in the top half, an odd rung, N.

IV.2 Clearing Lemma *(formerly the Zone Lemma — the parity audit showed it is not a patch on the odd-m witness but the escape engine itself, parity-free).* Suppose some square $s \neq m$ lies in the integer interval $(r, m + r]$ when $r \geq 1$, or in $(m, 2m)$ when $r = 0$. Then the entire hard class is N by the single move $-s$. Legality: $s \leq m + r$, the smallest hard member (for $r = 0$: $s < 2m$, and the smallest positive hard member is 2m), so the move is affordable to every member from the first. Landing: $n - s \equiv m + r - s \pmod{2m}$ is a foreign bottom — in (r, m) when $r < s < m$, in $[0, r)$ when $m < s \leq m + r$; for $r = 0$, the landing $2m - s$ lies in $(0, m)$. Combined with the free escape, class r is all N — except possibly $n = 0$, handled in IV.6.

IV.3 The interval always contains a square ($m \geq 3$) — repaired and parity-free. For $r \geq 1$: if $(r, m + r]$ contained no square, consecutive squares would straddle it, $c^2 \leq r$ and $(c + 1)^2 > m + r$, so $2c + 1 > m$, hence $c \geq m/2$ for any m, hence $c^2 \geq m^2/4 > m - 1 \geq r$, since $m^2/4 - (m - 1) = (m - 2)^2/4 > 0$ for $m \geq 3$ — contradicting $c^2 \leq r$. The inequality degenerates at exactly one modulus, $m = 2$, where $(m - 2)^2 = 0$ and the interval can indeed be empty ($(1, 3]$ in D(2, 1) contains no square). Erratum, folded in: for $r = 0$ the interval reads $(m, 2m)$, which contains a square for every $m \geq 3$ **except m = 4** — $(4, 8) = \{5, 6, 7\}$ is square-free, and it is the lone counterexample, since the same consecutive-squares squeeze forces emptiness only for $m \leq 5$, and $m = 3$ has its 4, $m = 5$ its 9.

IV.4 The failure window. With a square always present, the Clearing Lemma's hypothesis can fail for $r \geq 1$ only if the *unique* square in $(r, m + r]$ is m itself: $m = b^2$ with $(b - 1)^2 \leq r$ (no smaller square intrudes) and $(b + 1)^2 > m + r$, i.e. $r \leq 2b$. The window $(b - 1)^2 \leq r \leq 2b$ is nonempty iff $(b - 1)^2 \leq 2b$ iff $b \leq 3$. Among $m \geq 3$ that leaves exactly $m = 4$ ($b = 2$, $r \in \{1, 2, 3\}$) and $m = 9$ ($b = 3$, $r \in \{4, 5, 6\}$). With the $r = 0$ erratum, every class of $m = 4$ is condemned from the outside — no clearing square exists for any r — while $m = 9$ is healthy at every r except 4, 5, 6. Squarehood of m alone is harmless: at $m = 25$ ($b = 5$) the neighboring square 36 sits only $2b + 1 = 11$ above m, deep inside a 25-wide window, so it invades $(r, m + r]$ for every r; a square m isolates itself only when the modulus is small enough ($b \leq 3$) for square-gaps to outrun m.

IV.5 The rebels. At $(9, r)$ with $r \in \{4, 5, 6\}$, the smallest hard member $n_0 = m + r \in \{13, 14, 15\}$ can afford only the squares $k^2 \leq r$ and 9. A square $k^2 \leq r$ lands on $m + (r - k^2)$, the odd rung directly above bottom $r - k^2$ — N. The square 9 lands on r itself, the smallest free member — N, since it escapes by 1^2. Every move lands on N, so $\mathbf{n_0}$ **is P.** In-game receipts: in D(9,4), $13 \to 12, 9, 4$, all N; in D(9,5), $14 \to 13, 10, 5$, all N; in D(9,6), $15 \to 14, 11, 6$, all N. Above the rebel the class closes uniformly: the members $27 + r$ (31, 32, 33) are cleared by $25 \equiv 7 \pmod{18}$, which lies in all three hard sets, landing on bottoms 6, 7, 8; the members $45 + r$ (49, 50, 51) by $49 \equiv 13$, landing on bottoms 0, 1, 2; and from $63 + r \geq 64 = (m - 1)^2$ onward, the witness $64 \equiv 10 = m + 1$ lands on bottom $r - 1$ forever. So each exceptional game carries exactly one extra P-position, and the exception set is precisely the E displayed in the box.

IV.6 Corruption audits. The $n = 0$ clause: when $r = 0$, zero is claimed by squares mode and is the unique squares-mode position that can afford no square ($k \geq 1$ forces $k^2 \geq 1 > 0$), so it is terminal, hence P — while the unamended law would file residue 0 as N. It corrupts nobody: fallback chains never visit class 0; the positions that can drop to 0 are $n = k^2 \equiv 0 \pmod m$, class-0 members the theorem already declares N; and the classes above escape without it — spot receipts: $10 - 9 = 1$ (P) in D(5,0), $14 - 9 = 5$ (P) in D(7,0). For $r \geq 1$ the clause is absorbed, since 0 is itself a foreign bottom; it earns its keep exactly at $r = 0$. The rebels, one sentence in the same mirror: a rebel corrupts nobody either — fallback chains never enter class r, and the only positions with a move onto $m + r$ are class-r members above it (a square step back into class r forces $k^2 \equiv 0 \bmod 9$, i.e. $3 \mid k$), which the theorem already declares N, so the rebel's P-ness merely hands them one more witness for what they already were.

IV.7 Remarks. *(1) The odd-m witness identity, the original route.* For odd m, $(m - 1)^2 = m^2 - 2m + 1 \equiv m^2 + 1 \equiv$ **$m + 1 \pmod{2m}$**, using $m^2 \equiv m$ from Part III; and $m + 1$ belongs to every hard set, landing the hard residue exactly on bottom $r - 1$ ($m - 1$ when $r = 0$). So the pair $(1^2, (m - 1)^2)$ is a universal witness pair for all odd m at once — at the price of the threshold $n \geq (m - 1)^2$. The Clearing Lemma supersedes it: a square below $m + r$ does the same job with legality free, for both parities, with no threshold. The identity remains the cleanest line in this file and the historical door to everything above. *(2) The D(5,2) coincidence, resolved.* For any escaped game the P-set equals its window by construction: P-set = (chain P's) ∪ (class-r P's) = $T \cup \varnothing = T$. The equality fails exactly where escape does — the three exceptional games have P-set = $T \cup \{m + r\}$, and the diseased games' P-sets properly contain their windows.

Part V — Confinement and retrodiction

V.1 Confinement Lemma. For $m \in \{2, 4\}$ and every r: $Q(2m) \subseteq \{0, 1, m\}$ — read directly off the mod-4 and mod-8 rows of Part III — while every hard set H(m, r) excludes 0, 1, and m (IV.1). Hence from any hard-residue position, every legal square move either stays inside class r (a square $\equiv 0$ or $m \bmod 2m$ is $\equiv 0 \bmod m$) or lands one below on the odd rung above a bottom (a square $\equiv 1$), an N-position — never on a P-position of a foreign chain. *Proof:* only-options enumeration over Q(2m), the mirror image of the Independence section — there, fallback moves cannot leave a chain; here, square moves cannot leave the class except onto N. *Consequence:* the hard class's P/N structure is decided by a self-contained internal recursion, and its very first member already defies the chain law: 3 is P in D(2,1) and 5 is P in

D(4,1), where escape, if it existed, would force N. Calibration, as accepted at the viva: for $m = 2$ and $m = 4$ we prove confinement — no square door into the window, so the class turns inward. We do **not** prove that these confined classes obey no eventual law of their own: “no period and no modulus up to 200 fits” is a reading from the search instrument, not a theorem, and their aperiodicity remains open.

V.2 Double condemnation. The two diseased moduli are exactly the two the parity-free engine cannot serve. At $m = 2$ the engine is silent: the always-a-square inequality degenerates ($(m - 2)^2 = 0$) and the clearing interval $(1, 3]$ is genuinely square-free. At $m = 4$ the engine convicts every class from the outside: the failure window covers $r \in \{1, 2, 3\}$, and the $r = 0$ interval $(4, 8)$ is square-free. Outside view and inside view agree because both are reading the same two table rows: $Q(2m) \subseteq \{0, 1, m\}$.

V.3 Retrodiction I — the fingerprints. During the Chain Lemma verification sweeps — recorded weeks before any escape machinery existed — the ledger logged unexplained P-positions 3, 11, 23 in D(2,1) and 5, 13, 37 in D(4,1), anomalies with no mechanism attached. The Confinement Lemma now computes those games’ hard residues as 3 (mod 4) and 5 (mod 8), and all six fingerprints satisfy their congruence without exception. The theory did not accommodate old data; it specified coordinates for where lawlessness is possible, and data recorded before the theory existed turns out to have been sitting on those coordinates the entire time.

V.4 Retrodiction II — the D(9,4) receipt. Primary source, quoted verbatim from DIAGONAL_WORKBOOK.md and the raw JSON behind it:

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| D(9,4) | mod 18 | P ≡ 0,1,2,3,5,6,7,8 | preperiod 14 | period 18 | swept 200,000 |
```

{"game":[9,4],"preperiod":14,"period":18,"p_res":[0,1,2,3,5,6,7,8],"pre_P":[0,1,2,3,5,6,7,8,13]}

This line was recorded before the rebel mechanism existed, and the mechanism now derives every number on it with zero freedom. The rebel 13 is the lone off-residue P, named in the pre_P field. Any eventually-periodic fit beginning at or before 13 would propagate 13’s P-ness by the period, which the sweep refutes (31 is N); so every correct fit begins at 14, and the minimal preperiod is exactly rebel + 1 = 14. The period is $2m = 18$, and the P-residues are the foreign bottoms $\{0, \ldots, 8\}\ \{4\} = \{0, 1, 2, 3, 5, 6, 7, 8\}$. Provenance note, owed and recorded: an earlier session claimed from memory that every diagonal cell’s preperiod was ≤ 1 and built a certificate on that claim; the file said otherwise all along, the claim and the certificate are struck, and this line enters the paper as the primary-source receipt that replaced them.

Part VI — Scope

Solved: every $m \geq 3$ except $m = 4$ — all r, both parities — under the boxed law: the chains by the Chain Lemma, class r by the Clearing Lemma plus the free escape, the $n = 0$ clause at $r = 0$, and the three rebels of E; no fences, no thresholds, no machine dependence, and no even-m gap — the even case was never open, only misfiled while the Clearing Lemma wore the wrong name.

Explained: $m = 2$ and $m = 4$, where the Confinement Lemma proves escape impossible and hands the recorded lawlessness to the self-contained internal games — with the double condemnation of V.2 recording that the outside engine fails at exactly those two moduli and nowhere else.

Open: only the aperiodicity of the two confined classes; “lawless to 200,000, no modulus to 200” remains an instrument statement, not a theorem. The mod-16 and mod-20 stackings, once necessities for an even-m theorem, are demoted to cross-checks of a theorem already proved.